

School of Computer Science and Artificial Intelligence**Lab Assignment # 12.5**

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Submission Starts here**Screenshots:****Task Description #1 (Sorting – Merge Sort Implementation)****Prompt:**

create a function `merge_sort(arr)` that sorts a list in ascending order. include time complexity and space complexity in the function docstring. Verify the generated code with test cases.

Code:

```
1 def merge_sort(arr):
2     """
3     Sort a list in ascending order using the merge sort algorithm.
4
5     Time Complexity: O(n log n) - divides the list in half at each level (log n levels)
6     and merges n elements at each level.
7     Space Complexity: O(n) - additional temporary arrays used during merging;
8     recursion stack depth is O(log n) but dominant space is the
9     auxiliary arrays of size proportional to n.
10    """
11    if len(arr) <= 1:
12        return arr
13
14    mid = len(arr) // 2
15    left = merge_sort(arr[:mid])
16    right = merge_sort(arr[mid:])
17
18    return _merge(left, right)
19
20
21 def _merge(left, right):
22     """Merge two sorted lists into one sorted list."""
23     result = []
24     i = j = 0
25
26     while i < len(left) and j < len(right):
27         if left[i] <= right[j]:
28             result.append(left[i])
29             i += 1
30         else:
31             result.append(right[j])
32             j += 1
33
34     result.extend(left[i:])
35     result.extend(right[j:])
36     return result
37
38
39 if __name__ == "__main__":
40     # Test cases
41     test_cases = [
42         ([64, 34, 25, 12, 22, 11, 90], [11, 12, 22, 25, 34, 64, 90]),
43         ([5, 2, 8, 1, 9], [1, 2, 5, 8, 9]),
44         ([1], [1]),
45         ([], []),
46         ([3, 3, 3, 3], [3, 3, 3, 3]),
47         ([2, 1], [1, 2]),
48         ([5, 4, 3, 2, 1], [1, 2, 3, 4, 5]),
49     ]
50
51     passed = 0
52     for arr, expected in test_cases:
53         result = merge_sort(arr.copy())
54         status = "PASS" if result == expected else "FAIL"
55         if status == "PASS":
56             passed += 1
57         print(f"({status}): merge_sort({arr}) -> {result} (expected {expected})")
58
59     print(f"{passed}/{len(test_cases)} test cases passed.")
```

Output:

```
PASS: merge_sort([64, 34, 25, 12, 22, 11, 90]) -> [11, 12, 22, 25, 34, 64, 90] (expected [11, 12, 22, 25, 34, 64, 90])
PASS: merge_sort([5, 2, 8, 1, 9]) -> [1, 2, 5, 8, 9] (expected [1, 2, 5, 8, 9])
PASS: merge_sort([1]) -> [1] (expected [1])
PASS: merge_sort([]) -> [] (expected [])
PASS: merge_sort([3, 3, 3, 3]) -> [3, 3, 3, 3] (expected [3, 3, 3, 3])
PASS: merge_sort([2, 1]) -> [1, 2] (expected [1, 2])
PASS: merge_sort([64, 34, 25, 12, 22, 11, 90]) -> [11, 12, 22, 25, 34, 64, 90] (expected [11, 12, 22, 25, 34, 64, 90])
PASS: merge_sort([5, 2, 8, 1, 9]) -> [1, 2, 5, 8, 9] (expected [1, 2, 5, 8, 9])
PASS: merge_sort([1]) -> [1] (expected [1])
PASS: merge_sort([]) -> [] (expected [])
PASS: merge_sort([3, 3, 3, 3]) -> [3, 3, 3, 3] (expected [3, 3, 3, 3])
PASS: merge_sort([5, 2, 8, 1, 9]) -> [1, 2, 5, 8, 9] (expected [1, 2, 5, 8, 9])
PASS: merge_sort([1]) -> [1] (expected [1])
PASS: merge_sort([]) -> [] (expected [])
PASS: merge_sort([3, 3, 3, 3]) -> [3, 3, 3, 3] (expected [3, 3, 3, 3])
PASS: merge_sort([1]) -> [1] (expected [1])
PASS: merge_sort([3, 3, 3, 3]) -> [3, 3, 3, 3] (expected [3, 3, 3, 3])
PASS: merge_sort([3, 3, 3, 3]) -> [3, 3, 3, 3] (expected [3, 3, 3, 3])
PASS: merge_sort([2, 1]) -> [1, 2] (expected [1, 2])
PASS: merge_sort([5, 4, 3, 2, 1]) -> [1, 2, 3, 4, 5] (expected [1, 2, 3, 4, 5])
```

7/7 test cases passed.

Task Description #2 (Searching – Binary Search with AI)

Prompt:

create a function `binary_search(arr, target)` returning the index of the target or -1 if not found. Include docstrings explaining best, average, and worst-case complexities. Test with various inputs.

Code:

```
1 def binary_search(arr, target):
2     """
3     Time Complexity
4     -----
5     - Best case: O(1)
6       When the target is found at the middle position on the first comparison.
7     - Average case: O(log n)
8       On average, the search space is halved repeatedly until the target is
9       found or the search space is empty.
10    - Worst case: O(log n)
11      When the target is not present or is only found after reducing the
12      search space to size 1.
13
14    Space Complexity
15    -----
16    - O(1)
17      The algorithm uses a constant amount of extra space (iterative version).
18    """
19    left, right = 0, len(arr) - 1
20
21    while left <= right:
22        mid = (left + right) // 2
23        if arr[mid] == target:
24            return mid
25        elif arr[mid] < target:
26            left = mid + 1
27        else:
28            right = mid - 1
29
30    return -1
31
32
33 if __name__ == "__main__":
34     # Basic tests with integers
35     nums = [1, 3, 5, 7, 9, 11]
36     print("Searching in:", nums)
37     print("Target 1 -> index:", binary_search(nums, 1)) # best-case: first mid
38     print("Target 7 -> index:", binary_search(nums, 7)) # typical middle element
39     print("Target 11 -> index:", binary_search(nums, 11)) # last element
40     print("Target 4 -> index:", binary_search(nums, 4)) # not in list (should be -1)
41
42     # Tests with duplicates (binary search returns one valid index if present)
43     dup_nums = [2, 4, 4, 4, 6, 8]
44     print("\nSearching in (with duplicates):", dup_nums)
45     print("Target 4 -> index:", binary_search(dup_nums, 4))
46
47     # Tests with strings
48     words = ["apple", "banana", "cherry", "date", "fig", "grape"]
49     print("\nSearching in:", words)
50     print("Target 'cherry' -> index:", binary_search(words, "cherry"))
51     print("Target 'kiwi' -> index:", binary_search(words, "kiwi")) # not found
52
53     # Edge cases
54     empty = []
55     one_elem = [10]
56     print("\nEdge cases:")
57     print("Empty list, target 1 -> index:", binary_search(empty, 1))
58     print("Single element, target 10 -> index:", binary_search(one_elem, 10))
59     print("Single element, target 5 -> index:", binary_search(one_elem, 5))
```

Output:

```

Searching in: [1, 3, 5, 7, 9, 11]
Target 1 -> index: 0
Target 7 -> index: 3
Target 11 -> index: 5
Target 4 -> index: -1

Searching in (with duplicates): [2, 4, 4, 4, 6, 8]
Target 4 -> index: 2

Searching in: ['apple', 'banana', 'cherry', 'date', 'fig', 'grape']
Target 'cherry' -> index: 2
Target 'kiwi' -> index: -1

Edge cases:
Empty list, target 1 -> index: -1
Target 11 -> index: 5
Target 4 -> index: -1

Searching in (with duplicates): [2, 4, 4, 4, 6, 8]
Target 4 -> index: 2

Searching in: ['apple', 'banana', 'cherry', 'date', 'fig', 'grape']
Target 'cherry' -> index: 2
Target 'kiwi' -> index: -1

Edge cases:
Empty list, target 1 -> index: -1
Target 4 -> index: -1

Searching in (with duplicates): [2, 4, 4, 4, 6, 8]
Target 4 -> index: 2

Searching in: ['apple', 'banana', 'cherry', 'date', 'fig', 'grape']
Target 'cherry' -> index: 2
Target 'kiwi' -> index: -1

Edge cases:
Empty list, target 1 -> index: -1

Searching in (with duplicates): [2, 4, 4, 4, 6, 8]
Target 4 -> index: 2

Searching in: ['apple', 'banana', 'cherry', 'date', 'fig', 'grape']
Target 'cherry' -> index: 2
Target 'kiwi' -> index: -1

Edge cases:
Empty list, target 1 -> index: -1
Searching in (with duplicates): [2, 4, 4, 4, 6, 8]
Target 4 -> index: 2

Searching in: ['apple', 'banana', 'cherry', 'date', 'fig', 'grape']
Target 'cherry' -> index: 2
Target 'kiwi' -> index: -1

Edge cases:
Empty list, target 1 -> index: -1

Searching in: ['apple', 'banana', 'cherry', 'date', 'fig', 'grape']
Target 'cherry' -> index: 2
Target 'kiwi' -> index: -1

Edge cases:
Empty list, target 1 -> index: -1
Target 'cherry' -> index: 2
Target 'kiwi' -> index: -1

Edge cases:
Empty list, target 1 -> index: -1
Target 'kiwi' -> index: -1

Edge cases:
Empty list, target 1 -> index: -1
Edge cases:
Empty list, target 1 -> index: -1
Empty list, target 1 -> index: -1
Single element, target 10 -> index: 0
Single element, target 5 -> index: -1

```

Task Description #3: Smart Healthcare Appointment Scheduling System

Prompt:

A healthcare platform maintains appointment records containing appointment ID, patient name, doctor name, appointment time, and consultation fee. The system needs to:

1. Search appointments using appointment ID.
2. Sort appointments based on time or consultation fee.

Tasks

- recommend suitable searching and sorting algorithms.
- Justify the selected algorithms.
- Implement the algorithms in Python.

Code:

```
1 """
2 Healthcare Appointment System - Search and Sort Algorithms
3 =====
4 Appointment records: appointment ID, patient name, doctor name,
5 appointment time, consultation fee.
6
7 RECOMMENDATIONS & JUSTIFICATION:
8 =====
9 1. SEARCH BY APPOINTMENT ID: Binary Search
10 - Use when the list is sorted by appointment ID (or we maintain a sorted copy).
11 - Time: O(log n), Space: O(1) for iterative version.
12 - Justification: Appointment ID is a unique key; sorting by ID once allows
13   fast repeated lookups. Better than linear search O(n) when n is large.
14 - Alternative: Hash table (dict) for O(1) lookup if ID is the primary key.
15
16 2. SORT BY TIME OR FEE: Merge Sort
17 - Time: O(n log n) guaranteed; Space: O(n) for auxiliary array.
18 - Justification: Stable sort (preserves order of equal elements), predictable
19   performance, suitable for small to medium lists. Preferred when stability
20   matters (e.g., "same fee" order preserved by time).
21
22 """
23
24 from dataclasses import dataclass
25 from typing import Callable, Optional
26
27 @dataclass
28 class Appointment:
29     """Single appointment record."""
30     appointment_id: int
31     patient_name: str
32     doctor_name: str
33     appointment_time: str # e.g. "2025-02-28 10:30"
34     consultation_fee: float
35
36     def __repr__(self):
37         return (
38             f"Appointment(id={self.appointment_id}, patient={self.patient_name}, "
39             f"doctor={self.doctor_name}, time={self.appointment_time}, fee={self.consultation_fee})"
40         )
41
42 # =====
43 # 1. SEARCH BY APPOINTMENT ID (Binary Search)
44 #
45 # Precondition: appointments must be sorted by appointment_id (ascending).
46
47 def binary_search_by_id(
48     appointments: list[Appointment],
49     target_id: int,
50 ) -> Optional[Appointment]:
51     """
52     Search for an appointment by ID using binary search.
53     Returns the Appointment if found, else None.
54     """
55     if not appointments:
56         return None
57
58     # Ensure we search in a list sorted by appointment_id
59     sorted_by_id = sorted(appointments, key=lambda a: a.appointment_id)
60     left, right = 0, len(sorted_by_id) - 1
61     while left <= right:
62         mid = (left + right) // 2
63         a = sorted_by_id[mid]
64         if a.appointment_id == target_id:
65             return a
66         elif a.appointment_id < target_id:
67             left = mid + 1
68         else:
69             right = mid - 1
70     return None
71
72 # =====
73 # 2. SORT APPOINTMENTS (Merge Sort)
74 #
75 # Sort by a numeric or comparable key (e.g. time string, fee).
76 # Uses a key function: by_time or by_fee.
77
78 def merge_sort_appointments(
79     appointments: list[Appointment],
80     *,
81     by_time: bool = True,
82 ) -> list[Appointment]:
83     """
84     Sort appointments using Merge Sort.
85     by_time=True -> sort by appointment_time (string comparison).
86     """
```

```

87 by_time=False -> sort by consultation_fee (numeric).
88 Returns a new sorted list; does not modify the original.
89 Uses (key, appointment) pairs so the same merge logic works for any comparable key.
90 """
91 if not appointments:
92     return []
93 key_fn: Callable[[Appointment], str | float] = (
94     lambda a: a.appointment_time if by_time else (lambda a: a.consultation_fee)
95 )
96 keyed = [(key_fn(a), a) for a in appointments]
97 n = len(keyed)
98 temp: list[Optional[tuple]] = [None] * n
99
100 def merge_keyed(left: int, mid: int, right: int) -> None:
101     i, j, k = left, mid + 1, left
102     while i <= mid and j <= right:
103         if keyed[i][0] <= keyed[j][0]:
104             temp[k] = keyed[i]
105             i += 1
106         else:
107             temp[k] = keyed[j]
108             j += 1
109         k += 1
110     while i <= mid:
111         temp[k] = keyed[i]
112         i += 1
113         k += 1
114     while j <= right:
115         temp[k] = keyed[j]
116         j += 1
117         k += 1
118     for idx in range(left, right + 1):
119         keyed[idx] = temp[idx]
120
121 def merge_sort_keyed(l: int, r: int) -> None:
122     if l >= r:
123         return
124     m = (l + r) // 2
125     merge_sort_keyed(l, m)
126     merge_sort_keyed(m + 1, r)
127     merge_keyed(l, m, r)
128
129 merge_sort_keyed(0, n - 1)
130 return [keyed[i][1] for i in range(n)]
131
132
133 # -----
134 # Alternative: Merge sort with a simple key (no tuple) for clarity
135 # -----
136 def sort_appointments_by_time(appointments: list[Appointment]) -> list[Appointment]:
137     """Sort appointments by appointment time using Merge Sort."""
138     return merge_sort_appointments(appointments, by_time=True)
139
140
141 def sort_appointments_by_fee(appointments: list[Appointment]) -> list[Appointment]:
142     """Sort appointments by consultation fee using Merge Sort."""
143     return merge_sort_appointments(appointments, by_time=False)
144
145
146 # -----
147 # Demo
148 # -----
149 def main():
150     # Sample appointment records
151     appointments = [
152         Appointment(105, "Alice Chen", "Dr. Smith", "2025-02-20 14:00", 1500.0),
153         Appointment(102, "Bob Kim", "Dr. Jones", "2025-02-20 09:30", 2000.0),
154         Appointment(108, "Carol Lee", "Dr. Smith", "2025-02-20 11:00", 1500.0),
155         Appointment(101, "David Park", "Dr. Brown", "2025-02-20 08:00", 1200.0),
156         Appointment(110, "Eve Wong", "Dr. Jones", "2025-02-20 16:30", 2500.0),
157     ]
158
159     print("=== Healthcare Appointment System ===\n")
160     print("Original list (unsorted):")
161     for a in appointments:
162         print(f"  {a}")
163     print()
164
165     # 1. Search by appointment ID
166     print("---- Search by Appointment ID (Binary Search) ----")
167     for target in (102, 109):
168         result = binary_search_by_id(appointments, target)
169         print(f"  {target}: {result if result else 'Not found'}")
170     print()
171
172     # 2. Sort by time
173     print("---- Sorted by Appointment Time (Merge Sort) ----")
174     by_time = sort_appointments_by_time(appointments)
175     for a in by_time:
176         print(f"  {a} | {a.appointment_time} | {a.patient_name} | fee={a.consultation_fee}")
177     print()
178
179     # 3. Sort by consultation fee
180     print("---- Sorted by Consultation Fee (Merge Sort) ----")
181     by_fee = sort_appointments_by_fee(appointments)
182     for a in by_fee:
183         print(f"  {a} | fee={a.consultation_fee} | {a.patient_name} | {a.appointment_time}")
184
185
186 if __name__ == "__main__":
187     main()
188

```

Output:

```

=== Healthcare Appointment System ===

Original list (unsorted):
Appointment(id=105, patient='Alice Chen', doctor='Dr. Smith', time='2025-02-20 14:00', fee=1500.0)
Appointment(id=102, patient='Bob Kim', doctor='Dr. Jones', time='2025-02-20 09:30', fee=2000.0)
Appointment(id=108, patient='Carol Lee', doctor='Dr. Smith', time='2025-02-20 11:00', fee=1500.0)
Appointment(id=101, patient='David Park', doctor='Dr. Brown', time='2025-02-20 08:00', fee=1200.0)
Appointment(id=110, patient='Eve Wong', doctor='Dr. Jones', time='2025-02-20 16:30', fee=2500.0)

--- Search by Appointment ID (Binary Search) ---
ID 102: Appointment(id=102, patient='Bob Kim', doctor='Dr. Jones', time='2025-02-20 09:30', fee=2000.0)
ID 109: Not found

--- Sorted by Appointment Time (Merge Sort) ---
2025-02-20 08:00 | David Park | fee=1200.0
2025-02-20 09:30 | Bob Kim | fee=2000.0
2025-02-20 11:00 | Carol Lee | fee=1500.0
2025-02-20 14:00 | Alice Chen | fee=1500.0
2025-02-20 16:30 | Eve Wong | fee=2500.0

--- Sorted by Consultation Fee (Merge Sort) ---
fee=1200.0 | David Park | 2025-02-20 08:00
fee=1500.0 | Alice Chen | 2025-02-20 14:00
fee=1500.0 | Carol Lee | 2025-02-20 11:00
fee=2000.0 | Bob Kim | 2025-02-20 09:30
fee=2500.0 | Eve Wong | 2025-02-20 16:30

```


Task Description #4: Railway Ticket Reservation System Scenario

Prompt:

A railway reservation system stores booking details such as ticket ID, passenger name, train number, seat number, and travel date. The system must:

1. Search tickets using ticket ID.
2. Sort bookings based on travel date or seat number.

Tasks

- Identify efficient algorithms.
- Justify the algorithm choices.
- Implement searching and sorting in Python.

Code:

```

1  """
2  Railway Reservation System
3  Implements efficient search by ticket ID and sort by travel date or seat number.
4  """
5
6  from dataclasses import dataclass
7  from datetime import date
8  from typing import Optional
9
10
11 @dataclass
12 class Booking:
13     """Represents a single railway booking."""
14     ticket_id: str
15     passenger_name: str
16     train_number: str
17     seat_number: str
18     travel_date: date
19
20     def __str__(self):
21         return (
22             f"Ticket: {self.ticket_id} | {self.passenger_name} | "
23             f"Train: {self.train_number} | Seat: {self.seat_number} | "
24             f"Date: {self.travel_date}"
25         )
26
27
28 class RailwayReservationSystem:
29     """
30     Railway reservation system with:
31     - O(1) average-case search by ticket ID (hash table)
32     - O(n log n) sort by travel date or seat number (Timsort via sorted())
33     """
34
35     def __init__(self):
36         # list for ordered iteration and sorting
37         self._bookings: list[Booking] = []
38         # Hash table for O(1) lookup by ticket_id (algorithm choice: see search_ticket)
39         self._by_ticket_id: dict[str, Booking] = {}
40
41     def add_booking(self, booking: Booking) -> None:
42         """Add a booking. Duplicate ticket_id overwrites previous."""
43         self._by_ticket_id[booking.ticket_id] = booking
44         # Keep list in sync: remove old if same id, then append
45         self._bookings = [b for b in self._bookings if b.ticket_id != booking.ticket_id]
46         self._bookings.append(booking)
47
48     def search_ticket(self, ticket_id: str) -> Optional[Booking]:
49         """
50         Search by ticket ID.
51         Algorithm: Hash table (dict) lookup.
52         Justification: O(1) average-case lookup; ideal when the key is unique
53         (ticket ID). Binary search on sorted list would be O(log n) but requires
54         keeping the list sorted by ticket_id and O(log n) or O(n) insertions.
55         For "search by ID" as primary operation, hash table is the standard choice.
56         """
57         return self._by_ticket_id.get(ticket_id)
58
59     def sort_by_travel_date(self) -> list[Booking]:
60         """
61         Sort bookings by travel date (ascending).
62         Algorithm: Timsort (Python's sorted()).
63         Justification: O(n log n), stable sort. Stable means equal dates
64         keep their relative order, which is good for consistent display.
65         """
66         return sorted(self._bookings, key=lambda b: b.travel_date)
67
68     def sort_by_seat_number(self) -> list[Booking]:
69         """
70         Sort bookings by seat number (ascending).
71         Algorithm: Timsort (Python's sorted()).
72         Justification: O(n log n), stable sort. Stable means equal seat numbers
73         keep their relative order, which is good for consistent display.
74         """
75         return sorted(self._bookings, key=lambda b: b.seat_number)
76
77     def __str__(self):
78         return str(self._bookings)
79
80     def __len__(self):
81         return len(self._bookings)
82
83     def __iter__(self):
84         return iter(self._bookings)
85
86     def __getitem__(self, index: int) -> Booking:
87         return self._bookings[index]
88
89     def __setitem__(self, index: int, booking: Booking):
90         self._bookings[index] = booking
91         self._by_ticket_id[booking.ticket_id] = booking
92
93     def __delitem__(self, index: int):
94         del self._bookings[index]
95         ticket_id = self._bookings[index].ticket_id
96         del self._by_ticket_id[ticket_id]
97
98     def __contains__(self, ticket_id: str) -> bool:
99         return ticket_id in self._by_ticket_id
100
101     def __repr__(self):
102         return f"RailwayReservationSystem(bookings={self._bookings})"
103
104     def __eq__(self, other: RailwayReservationSystem) -> bool:
105         return self._bookings == other._bookings
106
107     def __ne__(self, other: RailwayReservationSystem) -> bool:
108         return self._bookings != other._bookings
109
110     def __lt__(self, other: RailwayReservationSystem) -> bool:
111         return len(self._bookings) < len(other._bookings)
112
113     def __le__(self, other: RailwayReservationSystem) -> bool:
114         return len(self._bookings) <= len(other._bookings)
115
116     def __gt__(self, other: RailwayReservationSystem) -> bool:
117         return len(self._bookings) > len(other._bookings)
118
119     def __ge__(self, other: RailwayReservationSystem) -> bool:
120         return len(self._bookings) >= len(other._bookings)
121
122     def __add__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
123         new_system = RailwayReservationSystem()
124         new_system._bookings = self._bookings + other._bookings
125         new_system._by_ticket_id = self._by_ticket_id | other._by_ticket_id
126         return new_system
127
128     def __sub__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
129         new_system = RailwayReservationSystem()
130         new_system._bookings = [b for b in self._bookings if b.ticket_id not in other._by_ticket_id]
131         new_system._by_ticket_id = {b.ticket_id: b for b in new_system._bookings}
132         return new_system
133
134     def __mul__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
135         new_system = RailwayReservationSystem()
136         new_system._bookings = self._bookings * other._bookings
137         new_system._by_ticket_id = self._by_ticket_id | other._by_ticket_id
138         return new_system
139
140     def __div__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
141         new_system = RailwayReservationSystem()
142         new_system._bookings = self._bookings // other._bookings
143         new_system._by_ticket_id = self._by_ticket_id | other._by_ticket_id
144         return new_system
145
146     def __mod__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
147         new_system = RailwayReservationSystem()
148         new_system._bookings = self._bookings % other._bookings
149         new_system._by_ticket_id = self._by_ticket_id | other._by_ticket_id
150         return new_system
151
152     def __pow__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
153         new_system = RailwayReservationSystem()
154         new_system._bookings = self._bookings ** other._bookings
155         new_system._by_ticket_id = self._by_ticket_id | other._by_ticket_id
156         return new_system
157
158     def __matmul__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
159         new_system = RailwayReservationSystem()
160         new_system._bookings = self._bookings * other._bookings
161         new_system._by_ticket_id = self._by_ticket_id | other._by_ticket_id
162         return new_system
163
164     def __divmod__(self, other: RailwayReservationSystem) -> (RailwayReservationSystem, RailwayReservationSystem):
165         new_system1 = RailwayReservationSystem()
166         new_system1._bookings = self._bookings // other._bookings
167         new_system1._by_ticket_id = self._by_ticket_id | other._by_ticket_id
168         new_system2 = RailwayReservationSystem()
169         new_system2._bookings = self._bookings % other._bookings
170         new_system2._by_ticket_id = self._by_ticket_id | other._by_ticket_id
171         return new_system1, new_system2
172
173     def __radd__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
174         new_system = RailwayReservationSystem()
175         new_system._bookings = other._bookings + self._bookings
176         new_system._by_ticket_id = other._by_ticket_id | self._by_ticket_id
177         return new_system
178
179     def __rsub__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
180         new_system = RailwayReservationSystem()
181         new_system._bookings = [b for b in other._bookings if b.ticket_id not in self._by_ticket_id]
182         new_system._by_ticket_id = {b.ticket_id: b for b in new_system._bookings}
183         return new_system
184
185     def __rmul__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
186         new_system = RailwayReservationSystem()
187         new_system._bookings = other._bookings * self._bookings
188         new_system._by_ticket_id = other._by_ticket_id | self._by_ticket_id
189         return new_system
190
191     def __rdiv__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
192         new_system = RailwayReservationSystem()
193         new_system._bookings = other._bookings // self._bookings
194         new_system._by_ticket_id = other._by_ticket_id | self._by_ticket_id
195         return new_system
196
197     def __rmod__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
198         new_system = RailwayReservationSystem()
199         new_system._bookings = other._bookings % self._bookings
200         new_system._by_ticket_id = other._by_ticket_id | self._by_ticket_id
201         return new_system
202
203     def __rpow__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
204         new_system = RailwayReservationSystem()
205         new_system._bookings = other._bookings ** self._bookings
206         new_system._by_ticket_id = other._by_ticket_id | self._by_ticket_id
207         return new_system
208
209     def __matmulr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
210         new_system = RailwayReservationSystem()
211         new_system._bookings = other._bookings * self._bookings
212         new_system._by_ticket_id = other._by_ticket_id | self._by_ticket_id
213         return new_system
214
215     def __divmodr__(self, other: RailwayReservationSystem) -> (RailwayReservationSystem, RailwayReservationSystem):
216         new_system1 = RailwayReservationSystem()
217         new_system1._bookings = other._bookings // self._bookings
218         new_system1._by_ticket_id = other._by_ticket_id | self._by_ticket_id
219         new_system2 = RailwayReservationSystem()
220         new_system2._bookings = other._bookings % self._bookings
221         new_system2._by_ticket_id = other._by_ticket_id | self._by_ticket_id
222         return new_system1, new_system2
223
224     def __iadd__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
225         self._bookings += other._bookings
226         self._by_ticket_id |= other._by_ticket_id
227         return self
228
229     def __isub__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
230         self._bookings = [b for b in self._bookings if b.ticket_id not in other._by_ticket_id]
231         self._by_ticket_id = {b.ticket_id: b for b in self._bookings}
232         return self
233
234     def __imul__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
235         self._bookings *= other._bookings
236         self._by_ticket_id |= other._by_ticket_id
237         return self
238
239     def __idiv__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
240         self._bookings //= other._bookings
241         self._by_ticket_id |= other._by_ticket_id
242         return self
243
244     def __imod__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
245         self._bookings %= other._bookings
246         self._by_ticket_id |= other._by_ticket_id
247         return self
248
249     def __ipow__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
250         self._bookings **= other._bookings
251         self._by_ticket_id |= other._by_ticket_id
252         return self
253
254     def __imatmul__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
255         self._bookings *= other._bookings
256         self._by_ticket_id |= other._by_ticket_id
257         return self
258
259     def __idivmod__(self, other: RailwayReservationSystem) -> (RailwayReservationSystem, RailwayReservationSystem):
260         new_system1 = RailwayReservationSystem()
261         new_system1._bookings = self._bookings // other._bookings
262         new_system1._by_ticket_id = self._by_ticket_id | other._by_ticket_id
263         new_system2 = RailwayReservationSystem()
264         new_system2._bookings = self._bookings % other._bookings
265         new_system2._by_ticket_id = self._by_ticket_id | other._by_ticket_id
266         return new_system1, new_system2
267
268     def __iaddr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
269         self._bookings += other._bookings
270         self._by_ticket_id |= other._by_ticket_id
271         return self
272
273     def __isubr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
274         self._bookings = [b for b in self._bookings if b.ticket_id not in other._by_ticket_id]
275         self._by_ticket_id = {b.ticket_id: b for b in self._bookings}
276         return self
277
278     def __imulr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
279         self._bookings *= other._bookings
280         self._by_ticket_id |= other._by_ticket_id
281         return self
282
283     def __idivr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
284         self._bookings //= other._bookings
285         self._by_ticket_id |= other._by_ticket_id
286         return self
287
288     def __imodr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
289         self._bookings %= other._bookings
290         self._by_ticket_id |= other._by_ticket_id
291         return self
292
293     def __ipowr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
294         self._bookings **= other._bookings
295         self._by_ticket_id |= other._by_ticket_id
296         return self
297
298     def __imatmulr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
299         self._bookings *= other._bookings
300         self._by_ticket_id |= other._by_ticket_id
301         return self
302
303     def __idivmodr__(self, other: RailwayReservationSystem) -> (RailwayReservationSystem, RailwayReservationSystem):
304         new_system1 = RailwayReservationSystem()
305         new_system1._bookings = self._bookings // other._bookings
306         new_system1._by_ticket_id = self._by_ticket_id | other._by_ticket_id
307         new_system2 = RailwayReservationSystem()
308         new_system2._bookings = self._bookings % other._bookings
309         new_system2._by_ticket_id = self._by_ticket_id | other._by_ticket_id
310         return new_system1, new_system2
311
312     def __iaddrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
313         self._bookings += other._bookings
314         self._by_ticket_id |= other._by_ticket_id
315         return self
316
317     def __isubrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
318         self._bookings = [b for b in self._bookings if b.ticket_id not in other._by_ticket_id]
319         self._by_ticket_id = {b.ticket_id: b for b in self._bookings}
320         return self
321
322     def __imulrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
323         self._bookings *= other._bookings
324         self._by_ticket_id |= other._by_ticket_id
325         return self
326
327     def __idivrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
328         self._bookings //= other._bookings
329         self._by_ticket_id |= other._by_ticket_id
330         return self
331
332     def __imodrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
333         self._bookings %= other._bookings
334         self._by_ticket_id |= other._by_ticket_id
335         return self
336
337     def __ipowrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
338         self._bookings **= other._bookings
339         self._by_ticket_id |= other._by_ticket_id
340         return self
341
342     def __imatmulrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
343         self._bookings *= other._bookings
344         self._by_ticket_id |= other._by_ticket_id
345         return self
346
347     def __idivmodrr__(self, other: RailwayReservationSystem) -> (RailwayReservationSystem, RailwayReservationSystem):
348         new_system1 = RailwayReservationSystem()
349         new_system1._bookings = self._bookings // other._bookings
350         new_system1._by_ticket_id = self._by_ticket_id | other._by_ticket_id
351         new_system2 = RailwayReservationSystem()
352         new_system2._bookings = self._bookings % other._bookings
353         new_system2._by_ticket_id = self._by_ticket_id | other._by_ticket_id
354         return new_system1, new_system2
355
356     def __iaddrrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
357         self._bookings += other._bookings
358         self._by_ticket_id |= other._by_ticket_id
359         return self
360
361     def __isubrrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
362         self._bookings = [b for b in self._bookings if b.ticket_id not in other._by_ticket_id]
363         self._by_ticket_id = {b.ticket_id: b for b in self._bookings}
364         return self
365
366     def __imulrrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
367         self._bookings *= other._bookings
368         self._by_ticket_id |= other._by_ticket_id
369         return self
370
371     def __idivrrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
372         self._bookings //= other._bookings
373         self._by_ticket_id |= other._by_ticket_id
374         return self
375
376     def __imodrrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
377         self._bookings %= other._bookings
378         self._by_ticket_id |= other._by_ticket_id
379         return self
380
381     def __ipowrrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
382         self._bookings **= other._bookings
383         self._by_ticket_id |= other._by_ticket_id
384         return self
385
386     def __imatmulrrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
387         self._bookings *= other._bookings
388         self._by_ticket_id |= other._by_ticket_id
389         return self
390
391     def __idivmodrrr__(self, other: RailwayReservationSystem) -> (RailwayReservationSystem, RailwayReservationSystem):
392         new_system1 = RailwayReservationSystem()
393         new_system1._bookings = self._bookings // other._bookings
394         new_system1._by_ticket_id = self._by_ticket_id | other._by_ticket_id
395         new_system2 = RailwayReservationSystem()
396         new_system2._bookings = self._bookings % other._bookings
397         new_system2._by_ticket_id = self._by_ticket_id | other._by_ticket_id
398         return new_system1, new_system2
399
400     def __iaddrrrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
401         self._bookings += other._bookings
402         self._by_ticket_id |= other._by_ticket_id
403         return self
404
405     def __isubrrrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
406         self._bookings = [b for b in self._bookings if b.ticket_id not in other._by_ticket_id]
407         self._by_ticket_id = {b.ticket_id: b for b in self._bookings}
408         return self
409
410     def __imulrrrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
411         self._bookings *= other._bookings
412         self._by_ticket_id |= other._by_ticket_id
413         return self
414
415     def __idivrrrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
416         self._bookings //= other._bookings
417         self._by_ticket_id |= other._by_ticket_id
418         return self
419
420     def __imodrrrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
421         self._bookings %= other._bookings
422         self._by_ticket_id |= other._by_ticket_id
423         return self
424
425     def __ipowrrrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
426         self._bookings **= other._bookings
427         self._by_ticket_id |= other._by_ticket_id
428         return self
429
430     def __imatmulrrrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
431         self._bookings *= other._bookings
432         self._by_ticket_id |= other._by_ticket_id
433         return self
434
435     def __idivmodrrrr__(self, other: RailwayReservationSystem) -> (RailwayReservationSystem, RailwayReservationSystem):
436         new_system1 = RailwayReservationSystem()
437         new_system1._bookings = self._bookings // other._bookings
438         new_system1._by_ticket_id = self._by_ticket_id | other._by_ticket_id
439         new_system2 = RailwayReservationSystem()
440         new_system2._bookings = self._bookings % other._bookings
441         new_system2._by_ticket_id = self._by_ticket_id | other._by_ticket_id
442         return new_system1, new_system2
443
444     def __iaddrrrrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
445         self._bookings += other._bookings
446         self._by_ticket_id |= other._by_ticket_id
447         return self
448
449     def __isubrrrrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
450         self._bookings = [b for b in self._bookings if b.ticket_id not in other._by_ticket_id]
451         self._by_ticket_id = {b.ticket_id: b for b in self._bookings}
452         return self
453
454     def __imulrrrrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
455         self._bookings *= other._bookings
456         self._by_ticket_id |= other._by_ticket_id
457         return self
458
459     def __idivrrrrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
460         self._bookings //= other._bookings
461         self._by_ticket_id |= other._by_ticket_id
462         return self
463
464     def __imodrrrrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
465         self._bookings %= other._bookings
466         self._by_ticket_id |= other._by_ticket_id
467         return self
468
469     def __ipowrrrrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
470         self._bookings **= other._bookings
471         self._by_ticket_id |= other._by_ticket_id
472         return self
473
474     def __imatmulrrrrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
475         self._bookings *= other._bookings
476         self._by_ticket_id |= other._by_ticket_id
477         return self
478
479     def __idivmodrrrrr__(self, other: RailwayReservationSystem) -> (RailwayReservationSystem, RailwayReservationSystem):
480         new_system1 = RailwayReservationSystem()
481         new_system1._bookings = self._bookings // other._bookings
482         new_system1._by_ticket_id = self._by_ticket_id | other._by_ticket_id
483         new_system2 = RailwayReservationSystem()
484         new_system2._bookings = self._bookings % other._bookings
485         new_system2._by_ticket_id = self._by_ticket_id | other._by_ticket_id
486         return new_system1, new_system2
487
488     def __iaddrrrrrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
489         self._bookings += other._bookings
490         self._by_ticket_id |= other._by_ticket_id
491         return self
492
493     def __isubrrrrrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
494         self._bookings = [b for b in self._bookings if b.ticket_id not in other._by_ticket_id]
495         self._by_ticket_id = {b.ticket_id: b for b in self._bookings}
496         return self
497
498     def __imulrrrrrr__(self, other: RailwayReservationSystem) -> RailwayReservationSystem:
499         self._bookings *= other._bookings
500         self._by_ticket_id |= other._by_ticket_id
501         return self
502
499

```

```

69         """
70         Sort bookings by seat number (ascending).
71         Algorithm: Timsort (Python's sorted()).
72         Justification: Same as above; we use key for natural ordering.
73         Seat numbers are compared as strings; for numeric ordering use a key
74         that parses to int if your format is purely numeric.
75         """
76         return sorted(self._bookings, key=lambda b: (b.seat_number, b.travel_date))
77
78     def get_all_bookings(self) -> list[Booking]:
79         """Return current list of bookings (unsorted)."""
80         return self._bookings.copy()
81
82
83 def main():
84     from datetime import date
85
86     system = RailwayReservationSystem()
87
88     # Sample bookings
89     system.add_booking(Booking("T001", "Alice", "TR-101", "A1", date(2025, 3, 15)))
90     system.add_booking(Booking("T002", "Bob", "TR-102", "B3", date(2025, 3, 10)))
91     system.add_booking(Booking("T003", "Carol", "TR-101", "A2", date(2025, 3, 20)))
92     system.add_booking(Booking("T004", "Dave", "TR-103", "C1", date(2025, 3, 12)))
93
94     print("=== Search by ticket ID ===")
95     for tid in ["T002", "T999"]:
96         b = system.search_ticket(tid)
97         print(f"{tid}: {b if b else 'Not found'}")
98
99     print("\n=== Sort by travel date ===")
100    for b in system.sort_by_travel_date():
101        print(f"{b}")
102
103    print("\n=== Sort by seat number ===")
104    for b in system.sort_by_seat_number():
105        print(f"{b}")
106
107
108 if __name__ == "__main__":
109     main()

```

Output:

```

=== Search by ticket ID ===
T002: Ticket: T002 | Bob | Train: TR-102 | Seat: B3 | Date: 2025-03-10
T999: Not found

=== Sort by travel date ===
Ticket: T002 | Bob | Train: TR-102 | Seat: B3 | Date: 2025-03-10
Ticket: T004 | Dave | Train: TR-103 | Seat: C1 | Date: 2025-03-12
Ticket: T001 | Alice | Train: TR-101 | Seat: A1 | Date: 2025-03-15
Ticket: T003 | Carol | Train: TR-101 | Seat: A2 | Date: 2025-03-20

=== Sort by seat number ===
Ticket: T001 | Alice | Train: TR-101 | Seat: A1 | Date: 2025-03-15
Ticket: T003 | Carol | Train: TR-101 | Seat: A2 | Date: 2025-03-20
Ticket: T002 | Bob | Train: TR-102 | Seat: B3 | Date: 2025-03-10
Ticket: T004 | Dave | Train: TR-103 | Seat: C1 | Date: 2025-03-12

```

Task Description #5: Smart Hostel Room Allocation System

Prompt:

A hostel management system stores student room allocation details including student ID, room number, floor, and allocation date. The system needs to:

1. Search allocation details using student ID.
2. Sort records based on room number or allocation date.

Tasks

- suggest optimized algorithms.
- Justify the selections.
- Implement the solution in Python.

Code:

```

1  """
2  Hostel Management System - Lab 2.2
3  Stores student room allocation details with optimized search and sort operations.
4  """
5
6  from dataclasses import dataclass
7  from datetime import date
8  from typing import Optional
9
10
11 @dataclass
12 class AllocationRecord:
13     """Single room allocation record."""
14     student_id: str
15     room_number: int
16     floor: int
17     allocation_date: date
18
19     def __str__(self):
20         return (
21             f"Student ID: {self.student_id}, Room: {self.room_number}, "
22             f"Floor: {self.floor}, Date: {self.allocation_date}"
23         )
24
25
26 class HostelManager:
27     """
28     Hostel allocation manager with:
29     - O(1) average search by student ID (hash table)
30     - O(n log n) sort by room number or allocation date (Timsort)
31     """
32
33     def __init__(self):
34         # Hash table: student_id -> AllocationRecord
35         # Chosen for O(1) average lookup when searching by student ID
36         self._by_student_id: dict[str, AllocationRecord] = {}
37         # List of all records (references same objects) for sorting
38         self._records: list[AllocationRecord] = []
39
40     def add_allocation(
41         self,
42         student_id: str,
43         room_number: int,
44         floor: int,
45         allocation_date: date,
46     ) -> bool:
47         """Add a new allocation. Returns False if student_id already allocated."""
48         if student_id in self._by_student_id:
49             return False
50         record = AllocationRecord(
51             student_id=student_id,
52             room_number=room_number,
53             floor=floor,
54             allocation_date=allocation_date,
55         )
56         self._by_student_id[student_id] = record
57         self._records.append(record)
58         return True
59
60     def search_by_student_id(self, student_id: str) -> Optional[AllocationRecord]:
61         """
62         Search allocation by student ID.
63         Algorithm: Hash table lookup.
64         Time: O(1) average, O(n) worst case (rare with good hash).
65         Justification: Student ID is unique; direct key access is optimal.
66         """
67         return self._by_student_id.get(student_id)
68
69     def get_sorted_by_room_number(self) -> list[AllocationRecord]:
70         """
71         Return all records sorted by room number.
72         Algorithm: Timsort (Python's sorted()).
73         Time: O(n log n). Stable sort preserves order for equal keys.
74         Justification: General-purpose, in-place style sort; no need for
75         maintaining a separate sorted structure when sort is on demand.
76         """
77         return sorted(self._records, key=lambda r: (r.room_number, r.floor))
78
79     def get_sorted_by_allocation_date(self) -> list[AllocationRecord]:
80         """
81         Return all records sorted by allocation date (ascending).
82         Algorithm: Timsort (Python's sorted()).
83         Time: O(n log n). Stable.
84         Justification: Same as room sort; date objects are comparable.
85         """
86         return sorted(self._records, key=lambda r: r.allocation_date)
87
88     def list_all(self) -> list[AllocationRecord]:
89         """Return all records in insertion order."""
90         return list[AllocationRecord](self._records)
91
92
93 # ----- Algorithm summary and justification -----
94 """
95 ALGORITHM CHOICES & JUSTIFICATION
96 -----
97
98 1. SEARCH BY STUDENT ID
99 - Chosen: Hash table (Python dict) keyed by student_id
100 - Time: O(1) average lookup
101 - Justification:
102   * Each student has at most one current allocation, so student_id is a
103   natural unique key.
104   * Hash table gives constant-time access by key without scanning.
105   * Alternatives rejected:
106     - Linear search O(n): poor for many records.
107     - Binary search O(log n): would require keeping list sorted by
108       student_id and extra bookkeeping when adding/removing.
109
110 2. SORT BY ROOM NUMBER / ALLOCATION DATE
111 - Chosen: Timsort via sorted(iterable, key=...)
112 - Time: O(n log n), stable
113 - Justification:
114   * We need to sort on different keys (room number, date) on demand.
115   * Timsort is Python's default; it is stable and efficient for
116   partially ordered data.
117   * Keeping two sorted copies (by room and by date) would give O(log n)
118   insertion but double storage and more complex updates; for typical
119   hostel size, sorting when needed is simpler and fast enough.
120 """
121

```



```

122
123 def main():
124     """Demo: add allocations, search by student ID, sort by room and date."""
125     from datetime import date
126
127     manager = HostelManager()
128
129     # Add sample allocations
130     manager.add_allocation("S001", 101, 1, date(2024, 1, 15))
131     manager.add_allocation("S002", 205, 2, date(2024, 2, 1))
132     manager.add_allocation("S003", 102, 1, date(2024, 1, 20))
133     manager.add_allocation("S004", 201, 2, date(2024, 1, 10))
134
135     print("--- Search by Student ID ---")
136     for sid in ["S002", "S999"]:
137         rec = manager.search_by_student_id(sid)
138         print(f" {sid}: {rec if rec else 'Not found'}")
139
140     print("\n--- Sorted by Room Number ---")
141     for r in manager.get_sorted_by_room_number():
142         print(f" {r}")
143
144     print("\n--- Sorted by Allocation Date ---")
145     for r in manager.get_sorted_by_allocation_date():
146         print(f" {r}")
147
148
149 if __name__ == "__main__":
150     main()

```

Output:

```

--- Search by Student ID ---
S002: Student ID: S002, Room: 205, Floor: 2, Date: 2024-02-01
S999: Not found

--- Sorted by Room Number ---
Student ID: S001, Room: 101, Floor: 1, Date: 2024-01-15
Student ID: S003, Room: 102, Floor: 1, Date: 2024-01-20
Student ID: S004, Room: 201, Floor: 2, Date: 2024-01-10
Student ID: S002, Room: 205, Floor: 2, Date: 2024-02-01

--- Sorted by Allocation Date ---
Student ID: S004, Room: 201, Floor: 2, Date: 2024-01-10
Student ID: S001, Room: 101, Floor: 1, Date: 2024-01-15
Student ID: S003, Room: 102, Floor: 1, Date: 2024-01-20
Student ID: S002, Room: 205, Floor: 2, Date: 2024-02-01

```

Task Description #6: Online Movie Streaming Platform

Prompt:

A streaming service maintains movie records with movie ID, title, genre, rating, and release year. The platform needs to:

1. Search movies by movie ID.
2. Sort movies based on rating or release year.

Tasks

- Recommend searching and sorting algorithms.
- Justify the chosen algorithms.
- Implement Python functions.

Code:

```

1  """
2  Streaming Service - Movie Records
3  Search by movie ID | Sort by rating or release year
4  """
5
6  from typing import List, Optional, Literal
7
8
9  # -----
10 # Algorithm recommendations & justification
11 # -----
12
13 SEARCH BY MOVIE ID:
14 Algorithm: Binary Search
15 Justification:
16 - Movie IDs are unique and comparable → sorted list by ID is well-defined.
17 - Time:  $O(\log n)$  per lookup vs  $O(n)$  for linear search → efficient for large catalogs.
18 - Space:  $O(1)$  extra; no need for a separate hash structure if we keep one list sorted by ID.
19 - Alternative: Hash table (dict) gives  $O(1)$  average lookup; use when ID lookups dominate
20   and the list is not already sorted by ID.
21
22 SORT BY RATING OR RELEASE YEAR:
23 Algorithm: Merge Sort
24 Justification:
25 - Stable sort: preserves order of movies with equal rating/year (e.g. by title or ID).
26 - Time:  $O(n \log n)$  in all cases; predictable for streaming-sized datasets.
27 - No worst-case  $O(n^2)$  as in naive Quick Sort; suitable for user-facing ordering.
28 - Works well with linked/record data; easy to sort by different keys (rating, year).
29 """
30
31
32 def binary_search_by_id(movies: List[dict], movie_id: int) -> Optional[dict]:
33     """
34     Search for a movie by ID using Binary Search.
35     Precondition: movies is sorted by 'id' (ascending).
36     Time:  $O(\log n)$ , Space:  $O(1)$ .
37     """
38     if not movies:
39         return None
40     left, right = 0, len(movies) - 1
41     while left <= right:
42         mid = (left + right) // 2
43         m = movies[mid]
44         if m["id"] == movie_id:
45             return m
46         if m["id"] < movie_id:
47             left = mid + 1
48         else:
49             right = mid - 1
50     return None
51
52
53 def merge(
54     arr: List[dict],
55     key: Literal["rating", "year", "id"],
56     ascending: bool,
57     left: int,
58     mid: int,
59     right: int,
60 ) -> None:
61     """Merge two sorted halves arr[left:mid+1] and arr[mid+1:right+1] in-place using temp buffer."""
62     left_copy = arr[left : mid + 1]
63     right_copy = arr[mid + 1 : right + 1]
64     i, j, k = 0, 0, left
65     while i < len(left_copy) and j < len(right_copy):
66         a_val = left_copy[i][key]
67         b_val = right_copy[j][key]
68         if ascending:
69             take_left = a_val <= b_val
70         else:
71             take_left = a_val >= b_val
72         if take_left:
73             arr[k] = left_copy[i]
74             i += 1
75         else:
76             arr[k] = right_copy[j]
77             j += 1
78         k += 1
79     while i < len(left_copy):
80         arr[k] = left_copy[i]
81         i, k = i + 1, k + 1
82     while j < len(right_copy):
83         arr[k] = right_copy[j]
84         j, k = j + 1, k + 1
85

```

```

86 def _merge_sort_range(
87     arr: List[dict],
88     key: Literal["rating", "year", "id"],
89     ascending: bool,
90     left: int,
91     right: int,
92 ) -> None:
93     """Recursive merge sort on arr[left:right+1] by key."""
94     if left >= right:
95         return
96     mid = (left + right) // 2
97     _merge_sort_range(arr, key, ascending, left, mid)
98     _merge_sort_range(arr, key, ascending, mid + 1, right)
99     _merge(arr, key, ascending, left, mid, right)
100
101
102
103 def sort_movies(
104     movies: List[dict],
105     by: Literal["rating", "year"],
106     ascending: bool = True,
107 ) -> List[dict]:
108     """
109     Sort movies by 'rating' or 'year' using Merge Sort (stable, O(n log n)).
110     Returns a new sorted list; does not mutate the original.
111     """
112     if not movies:
113         return []
114     result = [m.copy() for m in movies]
115     _merge_sort_range(result, by, ascending, 0, len(result) - 1)
116     return result
117
118
119 # -----
120 # Helpers: keep list sorted by ID for binary search; build from unsorted list
121 # -----
122 def sort_movies_by_id(movies: List[dict]) -> List[dict]:
123     """Sort by ID so binary_search_by_id can be used. Uses merge sort by 'id'."""
124     return sort_movies_by_key(movies, "id")
125
126
127 def sort_movies_by_key(
128     movies: List[dict],
129     key: Literal["rating", "year", "id"],
130     ascending: bool = True,
131 ) -> List[dict]:
132     """Generic merge sort by key ('id', 'rating', or 'year')."""
133     if not movies:
134         return []
135     result = [m.copy() for m in movies]
136     _merge_sort_range(result, key, ascending, 0, len(result) - 1)
137     return result
138
139
140 # -----
141 # Example usage
142 # -----
143 if __name__ == "__main__":
144     # Sample catalog (unsorted by ID)
145     catalog = [
146         {"id": 103, "title": "Inception", "genre": "Sci-Fi", "rating": 8.8, "year": 2010},
147         {"id": 101, "title": "The Shawshank Redemption", "genre": "Drama", "rating": 9.3, "year": 1994},
148         {"id": 102, "title": "The Dark Knight", "genre": "Action", "rating": 9.0, "year": 2008},
149         {"id": 105, "title": "Pulp Fiction", "genre": "Crime", "rating": 8.9, "year": 1994},
150         {"id": 104, "title": "Forrest Gump", "genre": "Drama", "rating": 8.8, "year": 1994},
151     ]
152
153     # 1) Sort by ID so we can use binary search
154     by_id = sort_movies_by_id(catalog)
155     print("Sorted by ID:", [m["id"] for m in by_id])
156
157     # 2) Search by movie ID
158     movie = binary_search_by_id(by_id, 102)
159     print("Search ID 102:", movie["title"] if movie else None)
160     print("Search ID 99:", binary_search_by_id(by_id, 99))
161
162     # 3) Sort by rating (descending = best first)
163     by_rating = sort_movies(catalog, "rating", ascending=False)
164     print("By rating (high first):", [(m["title"], m["rating"]) for m in by_rating])
165
166     # 4) Sort by release year (ascending)
167     by_year = sort_movies(catalog, "year", ascending=True)
168     print("By year (old first):", [(m["title"], m["year"]) for m in by_year])

```

Output:

```

Sorted by ID: [101, 102, 103, 104, 105]
Search ID 102: The Dark Knight
Search ID 99: None
By rating (high first): [('The Shawshank Redemption', 9.3), ('The Dark Knight', 9.0), ('Pulp Fiction', 8.9), ('Inception', 8.8), ('Forrest Gump', 8.8)]
By year (old first): [('The Shawshank Redemption', 1994), ('Pulp Fiction', 1994), ('Forrest Gump', 1994), ('The Dark Knight', 2008), ('Inception', 2010)]

```

Task Description #7: Smart Agriculture Crop Monitoring System

Prompt:

An agriculture monitoring system stores crop data with crop ID, crop name, soil moisture level, temperature, and yield estimate. Farmers need to:

1. Search crop details using crop ID.
2. Sort crops based on moisture level or yield estimate.

Tasks

- use reasoning to select algorithms.
- Justify algorithm suitability.
- Implement searching and sorting in Python.

Code:

```

1  """
2  Agriculture Monitoring System
3  Stores crop data and supports search by crop ID and sort by moisture or yield.
4  """
5
6  from dataclasses import dataclass
7  from typing import Optional
8
9
10 # -----
11 # Algorithm selection and justification
12 # -----
13 #
14 # SEARCH BY CROP ID:
15 # Chosen: Hash table (dict) index for O(1) lookup by crop_id.
16 # - Crop ID is unique and used for direct lookups.
17 # - Alternatives: linear search O(n), binary search O(log n) on sorted list.
18 # - Hash table is best here: constant-time access, no need to sort by ID
19 #   or scan the list. Suited when key is unique and lookups are frequent.
20 #
21 # SORT BY MOISTURE OR YIELD:
22 # Chosen: Python's sorted() (Timsort) with key function - O(n log n), stable.
23 # - We need to sort by different fields (moisture, yield) without changing
24 #   the original list order for display.
25 # - Timsort is efficient, stable (preserves order of equal elements), and
26 #   well-suited for real-world data. No need for manual quicksort/mergesort.
27 # -----
28
29
30 @dataclass
31 class Crop:
32     """Single crop record: id, name, soil moisture (%), temperature (°C), yield estimate."""
33     crop_id: str
34     name: str
35     soil_moisture: float
36     temperature: float
37     yield_estimate: float
38
39     def __str__(self):
40         return (
41             f"Crop(id={self.crop_id}, name={self.name!r}, "
42             f"moisture={self.soil_moisture}%, temp={self.temperature}°C, "
43             f"yield_est={self.yield_estimate})"
44         )
45
46
47 class AgricultureMonitoringSystem:
48     """
49     Manages crop data with O(1) search by crop ID and O(n log n) sort by
50     moisture or yield using Timsort (via sorted()).
51     """
52
53     def __init__(self):
54         self._crops: list[Crop] = []
55         self._by_id: dict[str, Crop] = {} # Hash index for search by ID
56
57     def add_crop(self, crop: Crop) -> None:
58         """Add a crop and update the ID index."""
59         self._crops.append(crop)
60         self._by_id[crop.crop_id] = crop
61
62     def search_by_id(self, crop_id: str) -> Optional[Crop]:
63         """
64         Search crop by ID using hash table lookup - O(1) average.
65         Returns the crop if found, else None.
66         """
67         return self._by_id.get(crop_id)
68
69     def sort_by_moisture(self, descending: bool = False) -> list[Crop]:
70         """

```

```

71     Return a new list of crops sorted by soil moisture.
72     Uses Timsort via sorted(); O(n log n), stable.
73     """
74     return sorted(
75         self._crops,
76         key=lambda c: c.soil_moisture,
77         reverse=descending,
78     )
79
80 def sort_by_yield_estimate(self, descending: bool = True) -> list[Crop]:
81     """
82     Return a new list of crops sorted by yield estimate.
83     Uses Timsort via sorted(); O(n log n), stable.
84     Default descending (highest yield first).
85     """
86     return sorted(
87         self._crops,
88         key=lambda c: c.yield_estimate,
89         reverse=descending,
90     )
91
92 def list_all(self) -> list[Crop]:
93     """Return current list of all crops (original order)."""
94     return list[Crop](self._crops)
95
96
97 def main():
98     system = AgricultureMonitoringSystem()
99
100     # Sample crop data
101     system.add_crop(Crop("C001", "Wheat", 45.2, 22.0, 3.8))
102     system.add_crop(Crop("C002", "Corn", 62.1, 25.5, 5.2))
103     system.add_crop(Crop("C003", "Rice", 78.0, 28.0, 4.1))
104     system.add_crop(Crop("C004", "Barley", 38.5, 20.0, 3.2))
105     system.add_crop(Crop("C005", "Soybean", 55.0, 24.0, 4.5))
106
107     print("=== Agriculture Monitoring System ===\n")
108
109     # 1. Search by crop ID
110     print("\n1. Search by crop ID")
111     print("-" * 40)
112     for cid in ["C002", "C009"]:
113         crop = system.search_by_id(cid)
114         if crop:
115             print(f"    Found: {crop}")
116         else:
117             print(f"    No crop with ID {cid!r}")
118
119     # 2. Sort by moisture (ascending: driest first)
120     print("\n2. Crops sorted by soil moisture (ascending)")
121     print("-" * 40)
122     for c in system.sort_by_moisture(descending=False):
123         print(f"    {c.soil_moisture:5.1f}% - {c.name} (ID: {c.crop_id})")
124
125     # 3. Sort by yield estimate (descending: highest first)
126     print("\n3. Crops sorted by yield estimate (descending)")
127     print("-" * 40)
128     for c in system.sort_by_yield_estimate(descending=True):
129         print(f"    {c.yield_estimate:4.1f} - {c.name} (ID: {c.crop_id})")
130
131     print("\nDone.")
132
133
134 if __name__ == "__main__":
135     main()

```


Output:

```
=== Agriculture Monitoring System ===

1. Search by crop ID
-----
Found: Crop(id=C002, name='Corn', moisture=62.1%, temp=25.5°C, yield_est=5.2)
No crop with ID 'C009'

2. Crops sorted by soil moisture (ascending)
-----
38.5% – Barley (ID: C004)
45.2% – Wheat (ID: C001)
55.0% – Soybean (ID: C005)
62.1% – Corn (ID: C002)
78.0% – Rice (ID: C003)

3. Crops sorted by yield estimate (descending)
-----
5.2 – Corn (ID: C002)
4.5 – Soybean (ID: C005)
4.1 – Rice (ID: C003)
3.8 – Wheat (ID: C001)
3.2 – Barley (ID: C004)

○ Done.
```

Task Description #8: Airport Flight Management System

Prompt:

An airport system stores flight information including flight ID, airline name, departure time, arrival time, and status. The system must:

1. Search flight details using flight ID.
2. Sort flights based on departure time or arrival time.

Tasks

- recommend algorithms.
- Justify the algorithm selection.
- Implement searching and sorting logic in Python.

Code:

```

1  """
2  Airport Flight Information System
3  - Search flight details by flight ID
4  - Sort flights by departure time or arrival time
5  """
6
7  from datetime import datetime
8  from typing import Optional
9
10
11  # ===== ALGORITHM RECOMMENDATIONS & JUSTIFICATION =====
12  #
13  # 1. SEARCH BY FLIGHT ID
14  # Recommended: Hash Table (Python dict) for  $O(1)$  average Lookup
15  # Justification:
16  # - Flight ID is a unique key; hash table gives constant-time lookup.
17  # - No need to keep list sorted by ID just for search.
18  # - Alternative: Binary Search  $O(\log n)$  if list were sorted by ID;
19  #   Linear Search  $O(n)$  is simple but slow for many flights.
20  #
21  # 2. SORT BY DEPARTURE/ARRIVAL TIME
22  # Recommended: Timsort (Python's sorted()) -  $O(n \log n)$ , stable
23  # Justification:
24  # - Timsort is Python's default; optimal for real-world data (handles
25  #   partial order, few comparisons).
26  # - Stable sort preserves relative order of equal keys (e.g. same time).
27  # - Alternatives: Merge Sort  $O(n \log n)$  stable; Quick Sort  $O(n \log n)$ 
28  #   average but not stable.
29  # =====
30
31
32  class Flight:
33      """Represents a single flight record."""
34
35      def __init__(
36          self,
37          flight_id: str,
38          airline: str,
39          departure_time: str,
40          arrival_time: str,
41          status: str,
42      ):
43          self.flight_id = flight_id
44          self.airline = airline
45          self.departure_time = departure_time # e.g. "14:30" or "2025-02-20 14:30"
46          self.arrival_time = arrival_time
47          self.status = status # e.g. "On Time", "Delayed", "Cancelled"
48
49      def __repr__(self):
50          return (
51              f"Flight(id={self.flight_id}, airline={self.airline}, "
52              f"dep={self.departure_time}, arr={self.arrival_time}, status={self.status})"
53          )
54
55
56  class AirportFlightSystem:
57      """
58      Manages flight data with:
59      -  $O(1)$  search by flight ID (hash table / dict)
60      -  $O(n \log n)$  sort by departure or arrival time (Timsort via sorted())
61      """
62
63      def __init__(self):
64          self._flights_list: list[Flight] = []
65          self._by_id: dict[str, Flight] = {} # Hash table for search by flight ID
66
67      def add_flight(self, flight: Flight) -> None:
68          """Add a flight; keeps list and index in sync."""
69          self._flights_list.append(flight)
70          self._by_id[flight.flight_id] = flight
71
72      def search_by_flight_id(self, flight_id: str) -> Optional[Flight]:
73          """
74          Search flight by ID using hash table lookup.
75          Algorithm: Hash table lookup -  $O(1)$  average time.
76          """
77          return self._by_id.get(flight_id)
78
79      def _parse_time(self, time_str: str) -> datetime:
80          """Parse time string for comparison. Supports 'HH:MM' or 'YYYY-MM-DD HH:MM'."""
81          time_str = time_str.strip()
82          for fmt in ("%Y-%m-%d %H:%M", "%H:%M", "%H:%M:%S"):
83              try:
84                  return datetime.strptime(time_str, fmt)
85              except ValueError:
86                  continue

```

```

87         raise ValueError(f"Cannot parse time: {time_str}")
88
89     def sort_by_departure_time(self, ascending: bool = True) -> list[Flight]:
90         """
91         Sort flights by departure time.
92         Algorithm: Timsort (sorted()) - O(n log n), stable.
93         """
94         return sorted(
95             self._flights_list,
96             key=lambda f: self._parse_time(f.departure_time),
97             reverse=not ascending,
98         )
99
100    def sort_by_arrival_time(self, ascending: bool = True) -> list[Flight]:
101        """
102        Sort flights by arrival time.
103        Algorithm: Timsort (sorted()) - O(n log n), stable.
104        """
105        return sorted(
106            self._flights_list,
107            key=lambda f: self._parse_time(f.arrival_time),
108            reverse=not ascending,
109        )
110
111    def get_all_flights(self) -> list[Flight]:
112        """Return current list of flights (unsorted)."""
113        return self._flights_list.copy()
114
115
116    # ===== DEMO / USAGE =====
117
118    def main():
119        system = AirportFlightSystem()
120
121        # Sample flights
122        flights_data = [
123            ("AA101", "American Airlines", "08:00", "11:30", "On Time"),
124            ("BA205", "British Airways", "14:30", "18:45", "Delayed"),
125            ("EK301", "Emirates", "06:15", "12:00", "On Time"),
126            ("LH402", "Lufthansa", "22:00", "02:30", "On Time"),
127            ("SQ501", "Singapore Airlines", "10:45", "16:20", "On Time"),
128        ]
129
130        for fid, airline, dep, arr, status in flights_data:
131            system.add_flight(Flight(fid, airline, dep, arr, status))
132
133        print("\n=== 1. SEARCH BY FLIGHT ID ===\n")
134        for fid in ["EK301", "XX999"]:
135            flight = system.search_by_flight_id(fid)
136            if flight:
137                print(f"Found: {flight}")
138            else:
139                print(f"Flight ID '{fid}' not found.")
140
141        print("\n=== 2. SORT BY DEPARTURE TIME (ascending) ===\n")
142        for f in system.sort_by_departure_time(ascending=True):
143            print(f" {f.departure_time} -> {f.flight_id} ({f.airline})")
144
145        print("\n=== 3. SORT BY ARRIVAL TIME (ascending) ===\n")
146        for f in system.sort_by_arrival_time(ascending=True):
147            print(f" {f.arrival_time} -> {f.flight_id} ({f.airline})")
148
149        print("\n=== 4. SORT BY DEPARTURE TIME (descending) ===\n")
150        for f in system.sort_by_departure_time(ascending=False):
151            print(f" {f.departure_time} -> {f.flight_id} ({f.airline})")
152
153
154    if __name__ == "__main__":
155        main()

```

Output:

```
=== 1. SEARCH BY FLIGHT ID ===

Found: Flight(id=EK301, airline=Emirates, dep=06:15, arr=12:00, status=On Time)
Flight ID 'XX999' not found.

=== 2. SORT BY DEPARTURE TIME (ascending) ===

06:15 -> EK301 (Emirates)
08:00 -> AA101 (American Airlines)
10:45 -> SQ501 (Singapore Airlines)
14:30 -> BA205 (British Airways)
22:00 -> LH402 (Lufthansa)

=== 3. SORT BY ARRIVAL TIME (ascending) ===

02:30 -> LH402 (Lufthansa)
11:30 -> AA101 (American Airlines)
12:00 -> EK301 (Emirates)
16:20 -> SQ501 (Singapore Airlines)
18:45 -> BA205 (British Airways)

=== 4. SORT BY DEPARTURE TIME (descending) ===

22:00 -> LH402 (Lufthansa)
14:30 -> BA205 (British Airways)
10:45 -> SQ501 (Singapore Airlines)
08:00 -> AA101 (American Airlines)
06:15 -> EK301 (Emirates)
```